Girwan Dhakal

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EDUCATION

The University of Alabama **Tuscaloosa, AL** GPA: 3.8 / 4
Bachelor of Science in Computer Science *Graduation Date: May 2027*
Master of Science in Computer Science

WORK EXPERIENCE

Alabama Credit Union **Tuscaloosa, AL**
Data Analyst Co-op *Aug 2025 - Present*

- Automated financial reporting using Microsoft Power Automate, cutting manual reporting time by ~40%.
- Created Power BI dashboards, increasing usage of dashboards for strategic decisions by ~50% and running Power BI training sessions for internal stakeholders.
- Built ad-hoc SQL queries on core banking data to support internal analysis and reporting.
- Optimized data-warehouse SQL scripts, cutting daily ETL update time by ~30%.

Alabama Life Research Institute **Tuscaloosa, AL**
Machine Learning Researcher *Jan 2025 - Present*

- First-authored peer-reviewed research paper on childhood speech development; collaborated with a cross-functional scientific team; manuscript awaiting publication.
- Developed classification models with PyTorch and Scikit-learn that achieved 84% accuracy in clinical population differentiation.
- Designed numerical embeddings for non-semantic speech and gesture data.
- Built scalable data preprocessing pipelines to parse behavioral transcript annotations into structured datasets; conducted exploratory data analysis and data visualizations in Jupyter to uncover patterns.
- Wrote an end-to-end multimodal ML pipeline combining speech and gesture modalities to detect joint attention behaviors in adults.
- Contribute to grant proposals by providing statistical analysis and proof-of-concept results that demonstrate project feasibility to funding agencies.

University of Alabama Libraries **Tuscaloosa, AL**
Software Developer Student Assistant *Jun 2025 - Aug 2025*

- Built and trained a PyTorch reinforcement learning model to personalize library resource recommendations for faculty, improving click-through rates by 15% and overall engagement by 50%.
- Authored tutorials on machine learning (ML) and High-Performance Computing (HPC) workflows, enabling 10+ faculty members with limited ML expertise to integrate HPC and machine learning in their research work.

PROJECTS

MONEYMENTOR BUDGETING TOOL

- Deployed a C# / ASP.NET Core full-stack budgeting application to Microsoft Azure, using Dockerized backend services, cloud-hosted RESTAPIs, and CI/CD pipelines.
- Designed and implemented REST APIs, authentication flows, and backend services with clean API contracts.
- Built responsive UI components in React with TypeScript, applying HTML/CSS fundamentals and accessibility-aware design patterns.

MULTI-THREADED GAME OF LIFE SIMULATION

- Built a high-performance C++ Game of Life simulation using MPI, multithreading, and CUDA C for parallel execution.
- Applied core computer science fundamentals including data structures, concurrency, memory management, and algorithmic optimization to improve program performance and scalability.
- Reduced execution time from 1 hour to 3 seconds through a CUDA-based shared memory design.

SKILLS

Languages: Python, C#, Java, C/C++ , JavaScript, SQL, Typescript
Frameworks: React, Next.js, MVC .NET Core, Django
Tools+DevOps: Docker, Kubernetes, Git, CI workflows, Distributed computing (MPI)

AWARDS

- 2nd Place UA Innovate Hackathon 2024
- University of Alabama Presidential and Engineering Scholarship
- Last Mile Education Fund Spring 2025 Scholarship